# Competitive Exam Practice 8: CPU Scheduling

GATE (Computer Science) Pattern

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October 29, 2026

Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type. Unless stated otherwise, ignore context-switch overhead, ties are broken in favour of the earlier arrival, and SRTF preempts only when a newcomer's burst is strictly shorter than the running process's remaining time. *All questions in this set are original, written in the GATE pattern; none is claimed to be a past-year question.*

**Q1** *[Original, GATE-pattern]* **[MCQ, 1 mark]** Consider the transitions (I) running → waiting, (II) running → ready, (III) waiting → ready, (IV) running → terminated. A *non-preemptive* CPU scheduler makes scheduling decisions only on:

(A) I and IV only (B) II and III only (C) I, II and IV only (D) I, II, III and IV

**Q2** *[Original, GATE-pattern]* **[MCQ, 1 mark]** For a set of processes that are all available at time 0, which policy is guaranteed to give the minimum average waiting time?

(A) FCFS (B) SJF (C) Longest-Job-First (D) Any policy, since total work is fixed

**Q3** *[Original, GATE-pattern]* **[MCQ, 1 mark]** Many I/O-bound processes repeatedly queue behind one CPU-bound process, leaving the I/O devices and then the CPU idle in turn. This behaviour, and the policy that suffers from it, are:

(A) starvation; SJF (B) convoy effect; FCFS (C) thrashing; FCFS (D) convoy effect; SRTF

**Q4** *[Original, GATE-pattern]* **[MCQ, 2 marks]** The next CPU burst is predicted by $\hat{\tau}_{n+1} = \alpha\tau_n + (1-\alpha)\hat{\tau}_n$. Which statement is **TRUE**?

(A) With $\alpha = 0$, the prediction equals the most recent measured burst

(B) With $\alpha = 1$, the prediction never changes from $\hat{\tau}_0$

(C) With $0 < \alpha < 1$, the weight on the burst $j$ steps back is $\alpha(1-\alpha)^j$

(D) The predictor makes SJF starvation-free

**Q5** *[Original, GATE-pattern]* **[NAT, 2 marks]** Four processes are scheduled FCFS:

| Process | Arrival time (ms) | CPU burst (ms) |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 1 | 3 |
| P3 | 3 | 6 |
| P4 | 5 | 1 |

Concepts That Click

The average waiting time (in ms) is _____.

**Q6** *[Original, GATE-pattern]* **[NAT, 2 marks]** Four processes are scheduled by non-preemptive SJF:

| Process | Arrival time (ms) | CPU burst (ms) |
|---|---|---|
| P1 | 0 | 8 |
| P2 | 2 | 5 |
| P3 | 3 | 2 |
| P4 | 4 | 3 |

The average turnaround time (in ms) is _____.

**Q7** *[Original, GATE-pattern]* **[NAT, 2 marks]** Four processes are scheduled by SRTF (preemptive SJF):

| Process | Arrival time (ms) | CPU burst (ms) |
|---|---|---|
| P1 | 0 | 10 |
| P2 | 2 | 3 |
| P3 | 3 | 1 |
| P4 | 6 | 4 |

The average waiting time (in ms) is _____.

**Q8** *[Original, GATE-pattern]* **[NAT, 1 mark]** For the SRTF schedule of Q7, the number of times the CPU is switched from one process to a *different* process (not counting the initial dispatch at time 0) is _____.

**Q9** *[Original, GATE-pattern]* **[NAT, 1 mark]** A scheduler uses exponential averaging with $\alpha = 0.5$ and initial guess $\hat{\tau}_0 = 12$ ms. The first two measured CPU bursts are 8 ms and 6 ms. The predicted length (in ms) of the third burst, $\hat{\tau}_2$, is _____.

**Q10** *[Original, GATE-pattern]* **[NAT, 2 marks]** Three processes with CPU bursts 10, 2 and 4 ms all arrive at time 0 and are run non-preemptively in some order. The difference between the largest and the smallest possible average waiting time (in ms, rounded to two decimal places) is _____.

Concepts That Click